

Grade 8
Unit 5
Lesson 9 Practice

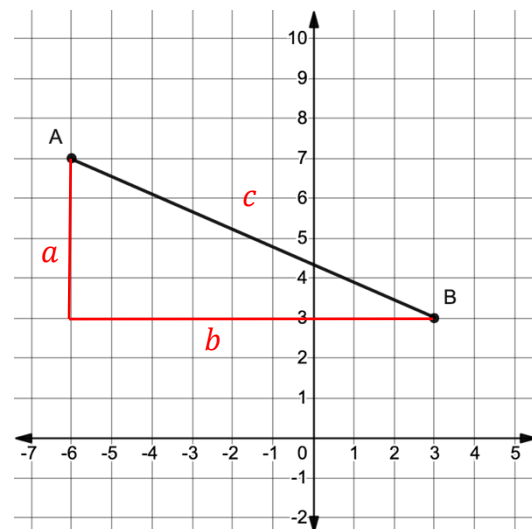
Name **Answer Key** _____
Date _____
Period _____

Use the graph for questions 1–2

On the graph label the line segment, c .

1. Create a right triangle with hypotenuse c by sketching horizontal and vertical legs. Label the legs of the right triangle a and b .
2. Use the Pythagorean Theorem to find the approximate length of c . Round to the nearest tenth.

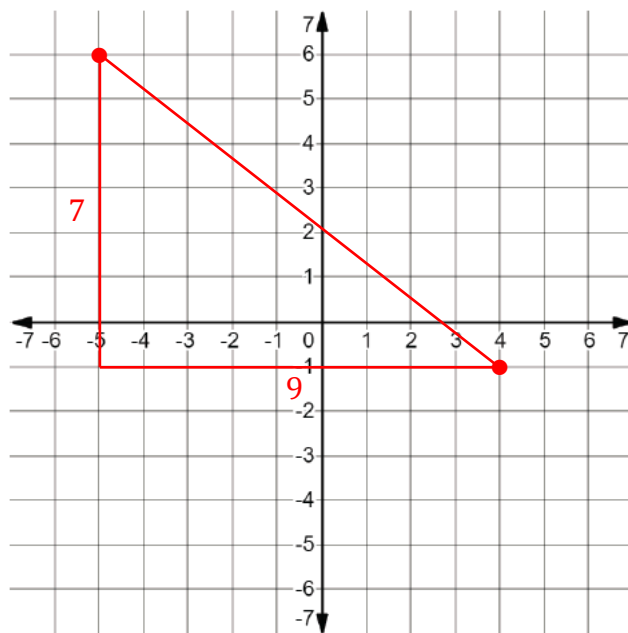
$$c^2 = 4^2 + 9^2 = 16 + 81 = 97$$
$$c = \sqrt{97} \approx 9.8 \text{ units}$$



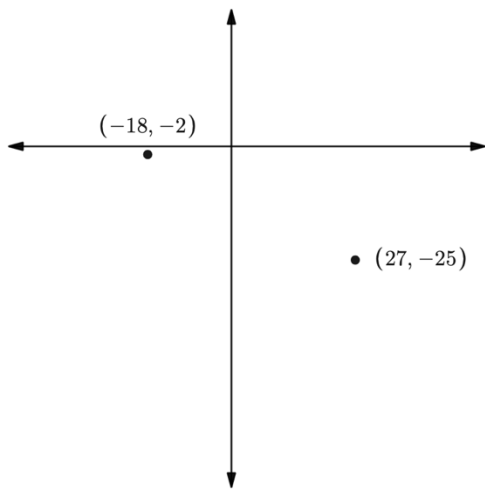
Use the coordinate grid below for questions 3–5.

3. Plot and connect the points $(4, -1)$ and $(-5, 6)$ on the grid.
4. Create a right triangle by drawing vertical and horizontal line segments on the grid.
5. Apply the Pythagorean Theorem to the exact distance between the points.

$$c^2 = 7^2 + 9^2$$
$$= 49 + 81$$
$$= 130$$
$$c = \sqrt{130} \approx 11.4$$



6. Find the exact distance between the points.



$$a = 23$$

$$b = 45$$

$$\begin{aligned} c^2 &= 23^2 + 45^2 \\ &= 529 + 2,025 \\ &= 2,554 \end{aligned}$$

$$\begin{aligned} c &= \sqrt{2,554} \\ c &\approx 50.5 \end{aligned}$$

Use the graph below of points A , B , and C for questions 7–10.

7. Find the length of $\overline{AB}$.

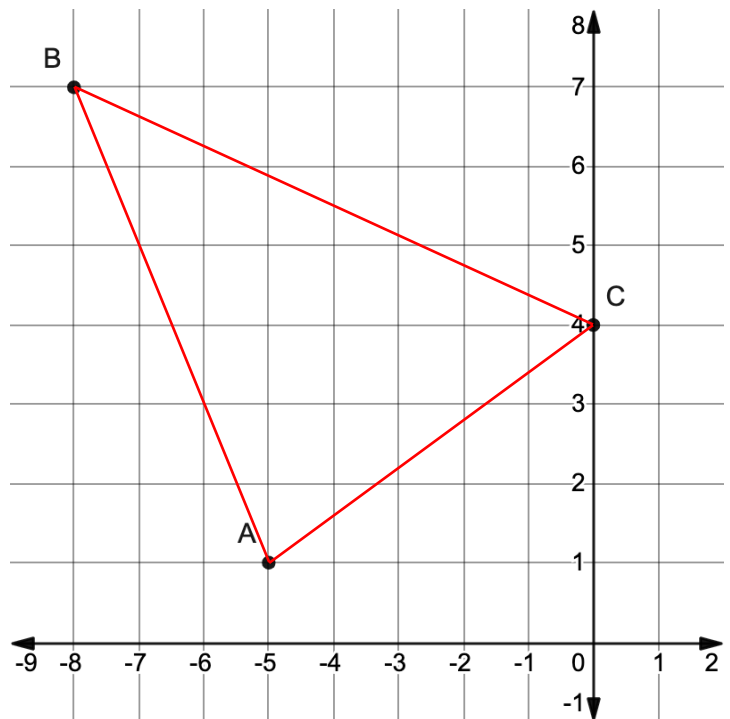
$$\begin{aligned} AB^2 &= 6^2 + 3^2 & AB &= \sqrt{45} \\ &= 36 + 9 & &\approx 6.7 \\ &= 45 \end{aligned}$$

8. Find exact length of $\overline{AC}$.

$$\begin{aligned} AC^2 &= 5^2 + 3^2 & AC &= \sqrt{34} \\ &= 25 + 9 & &\approx 5.8 \\ &= 34 \end{aligned}$$

9. Find exact length of $\overline{BC}$.

$$\begin{aligned} BC^2 &= 8^2 + 3^2 & BC &= \sqrt{73} \\ &= 64 + 9 & &\approx 8.5 \\ &= 73 \end{aligned}$$



10. Use the Converse of the Pythagorean Theorem to determine if Triangle ABC is a right triangle.

$$AB^2 + AC^2 \stackrel{?}{=} BC^2$$

$$45 + 34 \stackrel{?}{=} 73$$

$$79 \neq 73$$

Triangle ABC is not a right triangle.